Modeling Metallic Glass with Orbital Free – Density Functional Theory, OF-DFT

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Introduction

Metallic Glasses

Have no long-range order, are strong, corrosion-resistant, and conductive.

OF-DFT

OF-DFT scales ~O(N log N) vs. KS-DFT O(N^3).

Realistic System Sizes with OF-DFT

OF-DFT reaches scales that conventional DFT cannot.

Methods

System Setup

6×6×6 fcc Al supercell (a = 4.05 Å), 33% sites substituted to Mg → 216 atoms (Al-Mg eutectic)

DFT Method

Local LDA pseudopotentials (DFTpy .recpot) with the Wang-Teter (WT) kinetic energy density functional (KEDF), WT KEDF;

$T_{WT}[n] = T_{TF}[n] + T_{vW}[n] + T_{NL}[n]$

Job Submission

OF-DFT Molecular Dynamics (MD) on SDSC's Expanse cluster

Melt & Equilibrate

NPT dynamics (Berendsen thermostat + barostat, ambient pressure, 1 fs step): melted and held at 1,100 K

Stepped Quench

Discrete holds: 900 K → 700K → 500K → 300 K

Continuous Quench

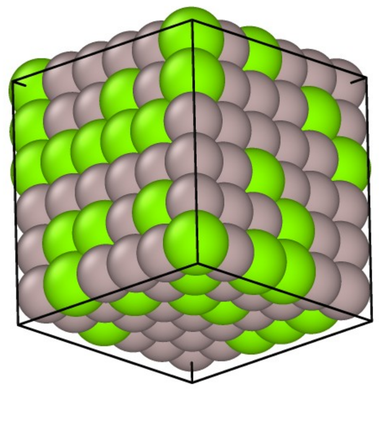
Linear cooling ramp to 300 K

Structural Analysis

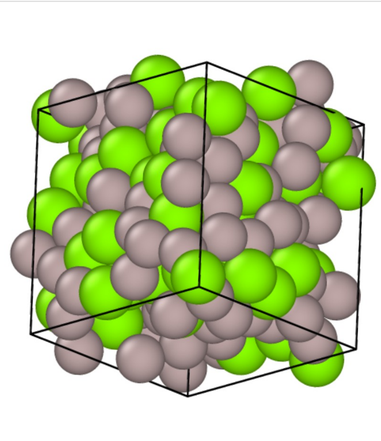
Structures visualized in OVITO; g(r) computed with a custom Python script

Results

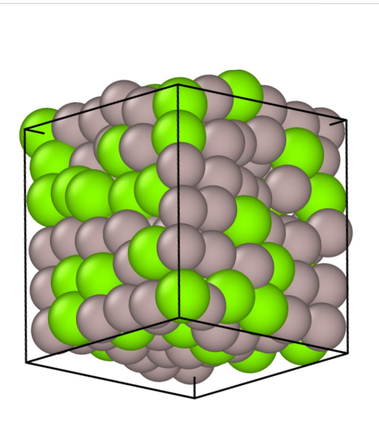
Two quench protocols — a stepped ramp and a continuous linear ramp — were run to completion on a 216-atom Al-Mg model system on SDSC Expanse. Both produced genuinely amorphous structures (below);



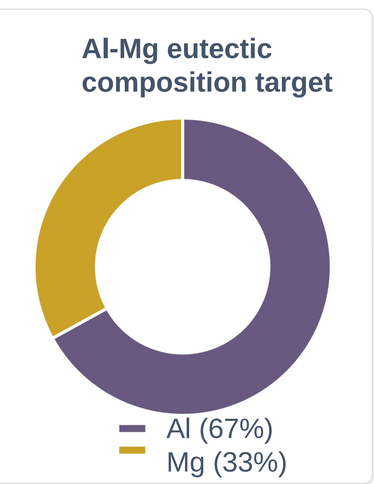
Crystalline starting structure



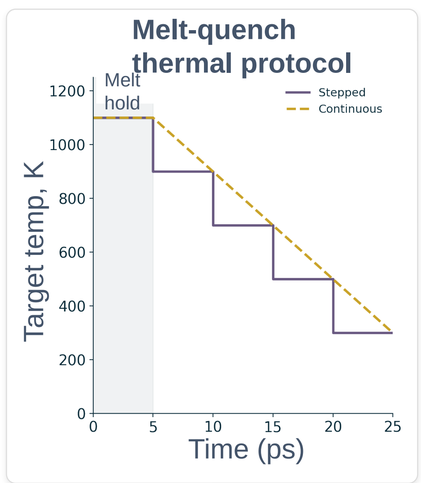
Melted structure 1,100 K



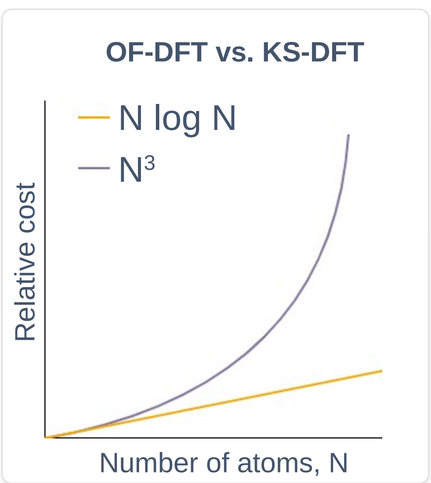
Quenched glass 300 K



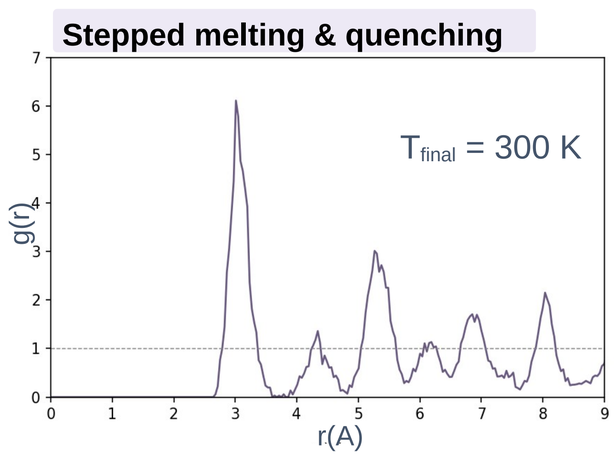
67:33 Al:Mg target composition

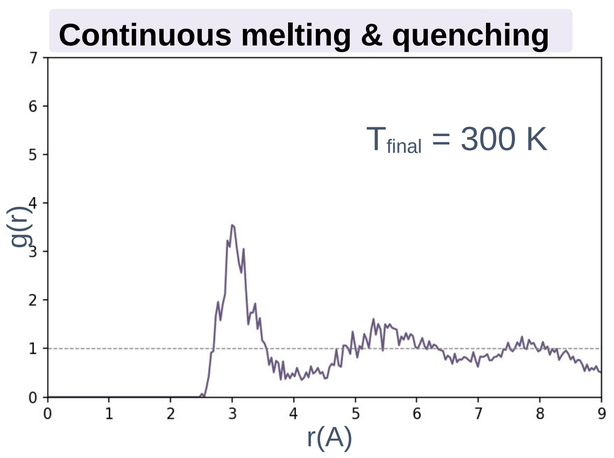


Stepped vs. continuous cooling ramps



OF-DFT scales far below KS-DFT cost





Amorphous Signature

Broad, split peaks match real amorphous Al-Mg alloys.

No Long-Range Order

No sharp peaks at larger r after the melt-quench.

Cost Efficiency at Scale

OF-DFT stays in orders of magnitude below KS-DFT beyond ~1,000 atoms.

Discussion

Amorphous Signature Confirmed

Broad, split g(r) peaks match real amorphous Al-Mg alloys.

Effect of Cooling Rate

The stepped process resulted in a crystal while the continuous process resulted into an amorphous glass: no long-range order detected.

Scaling to Production

216 atoms were simulated in Expanse.

Conclusions

Beyond KS-DFT's Reach

OF-DFT enables realistic, large-scale Al-Mg glass models & beyond.

Both stepped & continuous quenches successfully made a glass.

Future work

Extract Tg

Extract a reliable glass transition temperature (Tg); benchmark structure and thermal results against KS-DFT

New Compositions

Simulate larger super cells and extend the approach to other Metallic glasses.

Benchmarking

Compare OF-DFT with KS-DFT metallic glass simulations

Alfred University

References

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- San Diego Supercomputer Center. Expanse User Guide; UC San Diego, 2026.
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- Larsen, A. H., Mortensen, J. J., Blomqvist, J., Castelli, I. E., Christensen, R., Dulak, M., Friis, J., Groves, M., Hammer, B., ... Jacobsen, K. W. (2017). The Atomic Simulation Environment—a Python library for working with atoms. Journal of Physics: Condensed Matter, 29(27), 273002.

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